Graph Neural Networks for the Classification and Lead Identification of EGFR Inhibitors in Non-Small Cell Lung Cancer

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By

Krishna Gaire
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List of Abbreviations and Acronyms

ADMET	Absorption, Distribution, Metabolism, Excretion, and Toxicity
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
AUPRC	Area Under the Precision-Recall Curve
CADD	Computer-Aided Drug Design
ChEMBL	Chemistry European Molecular Biology Laboratory Database
ECFP	Extended Connectivity Fingerprint
EGFR	Epidermal Growth Factor Receptor
F1	F1-Score (Harmonic Mean of Precision and Recall)
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GIN	Graph Isomorphism Network
GNN	Graph Neural Network
GNNExplainer	Graph Neural Network Explainer
IC_{50}	Half-Maximal Inhibitory Concentration
IoST	Institute of Science and Technology
MCC	Matthews Correlation Coefficient
MPNN	Message Passing Neural Network
NSCLC	Non-Small Cell Lung Cancer
pIC_{50}	Negative Logarithm of IC_{50}
PyG	PyTorch Geometric
QED	Quantitative Estimate of Drug-likeness
RF	Random Forest
SMILES	Simplified Molecular Input Line Entry System
TKI	Tyrosine Kinase Inhibitor
TU	Tribhuvan University
VS	Virtual Screening

1. Introduction

1.1 Background

Cancer remains one of the most pressing public health challenges of the twenty-first century, imposing substantial clinical, economic, and societal burdens worldwide. Among all cancer types, lung cancer is the leading cause of cancer-related mortality. According to GLOBOCAN 2022 estimates, lung cancer accounted for approximately 1.8 million deaths globally in 2022, representing nearly 18% of all cancer deaths (Siegel et al., 2023). Non-Small Cell Lung Cancer (NSCLC) constitutes about 85% of all lung cancer cases. In Nepal and across South and East Asia, the burden of NSCLC is particularly acute. Most patients are diagnosed at advanced or metastatic stages due to limited screening infrastructure, resulting in poor prognosis and restricted access to precision medicine approaches (Siegel et al., 2023; Rotow & Jänne, 2023).

A major clinically validated therapeutic target in NSCLC is the Epidermal Growth Factor Receptor (EGFR), a transmembrane receptor tyrosine kinase belonging to the ErbB/HER family. Under normal physiological conditions, EGFR regulates cell growth, differentiation, and survival. Ligand binding (e.g., EGF or TGF-α) induces receptor dimerization, autophosphorylation of specific tyrosine residues, and activation of key downstream signalling cascades, including the RAS-RAF-MEK-ERK (MAPK) pathway, which promotes cell proliferation, and the PI3K-AKT-mTOR pathway, which enhances cell survival and inhibits apoptosis (Siegel et al., 2023; Rotow & Jänne, 2023).

In NSCLC, somatic mutations in the EGFR kinase domain cause ligand-independent constitutive activation, driving oncogenesis. The two predominant sensitizing mutations are in-frame deletions in exon 19 (Ex19del, approximately 45% of cases) and the L858R point mutation in exon 21 (approximately 40% of cases). These mutations collectively account for 85–90% of EGFR-mutant NSCLC. The prevalence of EGFR mutations varies significantly by ethnicity: 10–15% in Caucasian patients compared to 40–50% in East Asian populations. This ethnic disparity makes EGFR-targeted therapy especially relevant in the Nepali and broader South Asian clinical context, where late-stage presentation is common and access to advanced therapies remains limited (Rotow & Jänne, 2023; Leonetti et al., 2019).

Three generations of EGFR tyrosine kinase inhibitors (TKIs) have been developed to suppress this aberrant signalling. First-generation reversible inhibitors (gefitinib, erlotinib) and second-generation irreversible inhibitors (afatinib, dacomitinib) provided important clinical breakthroughs but were ultimately limited by acquired resistance, most notably the T790M gatekeeper mutation. Third-generation TKIs such as osimertinib were rationally designed to overcome T790M while sparing wild-type EGFR and have become the current standard of care. However, resistance to osimertinib typically emerges within 9–18 months, most commonly through the C797S mutation at the covalent binding site. This results in triple-mutant EGFR forms (e.g., Ex19del/T790M/C797S or L858R/T790M/C797S) that are resistant to all currently approved TKIs. This clinical reality creates an urgent unmet need for novel EGFR inhibitor chemotypes with activity against resistant mutants (Rotow & Jänne, 2023; Malik et al., 2025).

Traditional drug discovery pipelines are slow (often 10–15 years from discovery to approval), extremely costly (frequently exceeding USD 2 billion per approved drug), and suffer from high attrition rates. Computational approaches, especially machine learning, have emerged as powerful accelerators for large-scale *in silico* screening of chemical libraries (Schneider et al., 2020). Graph Neural Networks (GNNs) are particularly promising in this domain because molecules can be naturally represented as graphs, with atoms as nodes and chemical bonds as edges. Through iterative message-passing mechanisms, GNNs learn rich, data-driven molecular representations that frequently capture complex pharmacophore patterns more effectively than traditional hand-crafted fingerprints (Hu et al., 2020; Jiang et al., 2021).

Despite encouraging progress, significant gaps remain in the application of GNNs specifically to EGFR inhibitor discovery. Many existing models function as black-box systems, offering limited interpretability regarding which structural features drive activity predictions (Jiménez-Luna et al., 2020). Furthermore, few studies have extended high-performing GNN models beyond classification tasks to practical prospective virtual screening aimed at identifying novel lead compounds (Sun et al., 2021).

This thesis directly addresses these limitations by developing, optimising, and benchmarking advanced Graph Neural Network architectures — with a primary focus on AttentiveFP — for accurate binary classification of EGFR inhibitors, followed by a targeted virtual screening pipeline to prioritise potential novel leads.

1.2 Objectives

1.2.1 General Objective

The general objective of this study is to develop a robust Graph Neural Network-based computational framework for the binary classification of EGFR inhibitor activity (active versus inactive), rigorously evaluate its performance against traditional machine learning baselines, and deploy the optimised model in a virtual screening pipeline to identify, prioritise, and rank potential novel lead compounds from external chemical libraries for addressing drug-resistant NSCLC.

1.2.2 General Objective

The general objectives of this study are:

- Develop a robust Graph Neural Network based computational framework for the binary classification of EGFR inhibitor activity (active versus inactive).
- Evaluate and benchmark the framework's performance against traditional machine learning baselines to ensure predictive reliability.
- Use optimised model in a virtual screening pipeline to identify, prioritise, and rank potential lead compounds from external chemical libraries for addressing drug-resistant NSCLC.

1.3 Rationale

Clinical rationale: Acquired resistance to third-generation EGFR TKIs, particularly the emergence of triple mutants, represents a major unsolved problem in NSCLC management. No approved therapeutic effectively addresses the Ex19del/T790M/C797S mutant, leading to rapid disease progression after osimertinib failure (Rotow & Jänne, 2023).

Scientific rationale: Although GNNs have shown strong performance in molecular property prediction, their application to EGFR inhibitors has been constrained by methodological shortcomings such as random splitting, insufficient architectural comparison, limited featurisation, and lack of interpretability (Malik et al., 2025; Jiang et al., 2021; Jiménez-Luna et al., 2020). This thesis bridges these gaps through scaffold-aware evaluation, rich graph features, Optuna optimisation, GNNExplainer, and prospective virtual screening.

Methodological and practical rationale: The project uses only open-source tools (RDKit, PyTorch Geometric, Optuna) and publicly available ChEMBL data, ensuring full reproducibility. The scope is appropriate for a Master's thesis — computationally feasible while producing both methodological contributions and a ranked list of potential lead candidates.

1.4 Research Questions

1. To what extent do advanced Graph Neural Network architectures, particularly AttentiveFP with comprehensive node and edge features, outperform or match conventional machine learning baselines (Random Forest and Logistic Regression) in binary EGFR inhibitor classification under rigorous Bemis-Murcko scaffold splitting?

2. How effectively can the optimised AttentiveFP model, supported by GNNExplainer interpretability, be applied in a targeted virtual screening pipeline to identify, prioritise, and rank novel potential EGFR inhibitors from an external library of related-kinase compounds?

1.5 Significance of the Study

This research makes several important contributions to computational chemistry and drug discovery. First, it provides a methodologically rigorous evaluation of multiple GNN architectures with rich atom- and bond-level features and Bayesian hyperparameter optimisation. Second, it delivers systematic instance-level interpretability using GNNExplainer, connecting model predictions to established EGFR pharmacophores and offering chemically actionable insights. Third, by applying the best model to virtual screening of unseen compounds from related kinases, the study generates a prioritised list of potential novel lead candidates with favourable drug-likeness profiles. The work is also regionally relevant for Nepal, where the NSCLC burden is rising and access to next-generation therapies is limited.

1.6 Limitations

- **Computational only:** No experimental (in vitro or in vivo) validation of predicted leads was performed.
- **Data dependence:** Model performance is limited by the quality and heterogeneity of ChEMBL bioactivity data.
- **2D graph representation:** Three-dimensional conformational effects and protein-ligand binding geometry are not explicitly modelled.
- **Binary classification:** The study focuses on active/inactive classification; quantitative pIC_{50} regression is outside the current scope.
- **Wild-type EGFR focus:** The models target wild-type EGFR (CHEMBL203); performance on mutant forms requires future work.
- **Virtual screening predictions:** All prioritised compounds remain computational hypotheses requiring experimental confirmation.

2. Literature Review

2.1 EGFR as a Therapeutic Target in NSCLC

The Epidermal Growth Factor Receptor (EGFR), also known as ErbB1 or HER1, is a transmembrane glycoprotein and a member of the ErbB family of receptor tyrosine kinases (RTKs) (Siegel et al. 2023). Structurally, EGFR comprises an extracellular ligand-binding domain, a single transmembrane helix, an intracellular juxtamembrane region, a tyrosine kinase domain, and a C-terminal regulatory tail. Under physiological conditions, ligand binding (e.g., epidermal growth factor [EGF] or transforming growth factor-alpha [TGF-α]) induces receptor dimerization, autophosphorylation of specific tyrosine residues, and subsequent activation of multiple downstream signaling pathways. The primary pathways include the RAS-RAF-MEK-ERK (MAPK) pathway, which promotes cell proliferation, and the PI3K-AKT-mTOR pathway, which enhances cell survival and inhibits apoptosis (Siegel et al. 2023; Rotow and Jänne 2023).

In non-small cell lung cancer (NSCLC), activating somatic mutations in the EGFR kinase domain lead to ligand-independent constitutive signaling, driving oncogenesis. The two most prevalent sensitizing mutations are in-frame deletions in exon 19 (Ex19del, approximately 45% of cases) and the L858R point mutation in exon 21 (approximately 40% of cases). Together, these mutations account for 85–90% of EGFR-mutant NSCLC (Rotow and Jänne 2023). The overall prevalence of EGFR mutations in NSCLC varies by ethnicity: 10–15% in Caucasian populations and 40–50% in East Asian populations (Siegel et al. 2023). This ethnic disparity is particularly relevant in regions such as Nepal and South Asia, where NSCLC is often diagnosed at advanced stages and access to molecular testing and targeted therapies remains limited. Consequently, EGFR has become one of the most important therapeutic targets in precision oncology for NSCLC, with EGFR tyrosine kinase inhibitors (TKIs) transforming patient outcomes in mutation-positive cases (Passaro et al. 2021).

2.2 Generations of EGFR TKIs and Resistance Mechanisms

The evolution of EGFR TKIs reflects an iterative response to emerging resistance mechanisms, progressing through three distinct generations (Rotow and Jänne 2023).

First-generation TKIs, such as gefitinib and erlotinib, are reversible, ATP-competitive inhibitors that bind preferentially to mutant EGFR. These agents produced unprecedented response rates and progression-free survival benefits in patients with sensitizing mutations. However, acquired resistance developed in nearly all patients within 9–13 months, predominantly through the secondary T790M "gatekeeper" mutation in exon

20. This substitution increases ATP affinity while sterically hindering drug binding (Passaro et al. 2021).

Second-generation TKIs, including afatinib and dacomitinib, are irreversible covalent inhibitors targeting Cys797 in the ATP-binding pocket. They exhibit broader activity against T790M in preclinical models and inhibit other ErbB family members. Despite improved potency, their clinical use has been restricted by on-target toxicity to wild-type EGFR, leading to dose-limiting adverse effects such as skin rash and diarrhea (Passaro et al. 2021; Rotow and Jänne 2023).

Third-generation TKIs, most notably osimertinib, were rationally designed to covalently bind T790M-mutant EGFR while sparing wild-type EGFR. Osimertinib is currently the preferred first-line therapy for EGFR-mutant NSCLC, demonstrating superior efficacy and central nervous system penetration compared to earlier generations. Nevertheless, resistance inevitably emerges, most commonly via the C797S mutation at the covalent binding site. This leads to triple-mutant EGFR (e.g., Ex19del/T790M/C797S or L858R/T790M/C797S), which is resistant to all approved TKIs and represents a major unmet clinical need in advanced NSCLC treatment (Rotow and Jänne 2023; Malik, Sharma, and Kumar 2025).

To address this critical gap, fourth-generation EGFR TKIs, such as BLU-945, are currently undergoing clinical evaluation. These novel agents are specifically designed as reversible, wild-type-sparing inhibitors that maintain robust *in vivo* potency against the triple-mutant configurations that mediate third-generation resistance (Lim et al. 2024).

2.3 Bioactivity Data and the ChEMBL Database

The ChEMBL database, maintained by the European Molecular Biology Laboratory-European Bioinformatics Institute (EMBL-EBI), is one of the largest and most widely utilised public repositories of bioactive molecules. Following recent comprehensive updates, ChEMBL now contains over 2.3 million distinct compounds and more than 20 million bioactivity measurements across thousands of biological targets (Zdrazil et al. 2024). For EGFR research, the relevant entry is target ID CHEMBL203, which aggregates bioactivity data from diverse published and deposited sources (Zdrazil et al. 2024).

A major challenge when utilising ChEMBL data for machine learning is the inherent heterogeneity of experimental conditions. Bioactivity values (primarily IC_{50}) originate from hundreds of different laboratories employing varied assay protocols, cell lines, and conditions. To mitigate this, rigorous, modern curation practices mandate strict filtering for exact IC_{50} values, conversion to the logarithmic pIC_{50} scale, removal of ambiguous data points, and aggregation of multiple measurements per compound (Huang et al. 2021). In the present study, additional standardisation using RDKit (salt removal, canonicalisation, and molecular weight filtering between 100–1000 Da) yielded a high-quality dataset of 10,523 unique compounds, with approximately 74% classified as active at the commonly used threshold of $pIC_{50} \geq 6.0$ (equivalent to $IC_{50} \leq 1\ \mu M$) (Huang et al. 2021; Malik, Sharma, and Kumar 2025).

2.4 Molecular Representation and Traditional Machine Learning Approaches

Prior to deep learning approaches, molecular property prediction relied heavily on fixed-length molecular fingerprints paired with classical machine learning algorithms. The Extended Connectivity Fingerprint (ECFP), which encodes local atomic environments by iteratively capturing circular neighbourhoods, has long served as a de facto baseline standard in computational cheminformatics (Deng et al. 2022). ECFP4 (typically folded to 2048 bits) provides a balance of computational efficiency and chemical informativeness.

These fingerprints are commonly combined with ensemble methods such as Random Forest (RF) or simpler linear models like Logistic Regression. Random Forest constructs multiple decision trees on bootstrapped data subsets and aggregates predictions, providing robustness to noise and intrinsic feature importance rankings (Jiang et al. 2021). Logistic Regression, while less complex, offers high interpretability through coefficient analysis. Despite their widespread adoption and strong baseline performance, these approaches suffer from a fundamental limitation: the molecular representation is hand-crafted and static, potentially missing higher-order topological patterns and long-range atomic interactions critical to biological activity (Jiang et al. 2021; Deng et al. 2022).

2.5 Graph Neural Networks for Molecular Property Prediction

Graph Neural Networks (GNNs) have emerged as a transformative approach in molecular machine learning by learning representations directly from the molecular graph structure rather than relying on pre-defined fingerprints (Gilmer et al. 2017). In the message-passing framework, each atom (node) iteratively aggregates features from its neighbouring atoms (edges) across multiple layers. This process enables the model to encode increasingly complex chemical contexts. After the message-passing phase, a readout function produces a fixed-size graph-level embedding suitable for classification tasks (Gilmer et al. 2017).

Prominent GNN architectures evaluated in recent drug discovery frameworks include the Graph Convolutional Network (GCN), which performs degree-normalised aggregation, and the Graph Attention Network (GAT), which introduces learnable attention coefficients (Y. Zhang et al. 2021). More advanced architectures, such as AttentiveFP, employ multi-step attentive pooling and natively incorporate edge features, demonstrating particular strength in capturing local and global pharmacophore information through hierarchical attention mechanisms (Xiong et al. 2020; Jiang et al. 2021).

Large-scale studies have confirmed that GNNs frequently outperform traditional fingerprint-based methods on structurally diverse datasets, and benefit significantly from self-supervised pre-training on millions of unlabelled molecules (Hu et al. 2020). Recently, the field has also seen a rapid expansion into 3D geometric deep learning. Geometric GNNs explicitly incorporate three-dimensional atomic coordinates and spatial symmetries (e.g., rotational equivariance), offering more expressive representations of protein-ligand binding conformations than standard 2D topological graphs (Atz, Grisoni, and G. Schneider 2021).

2.6 GNN Models for EGFR Inhibitor Prediction

One of the most relevant prior works is DeepEGFR, a GAT-based model trained on ChEMBL-derived EGFR bioactivity data (Malik, Sharma, and Kumar 2025). DeepEGFR achieved strong classification performance in distinguishing active from inactive EGFR inhibitors. However, several limitations remain: reliance on random data splitting (which often overestimates generalisation to novel scaffolds), absence of instance-level structural explanations, and lack of prospective application to virtual screening for new lead compounds (Malik, Sharma, and Kumar 2025; Jiang et al. 2021).

The present thesis extends this line of research by systematically evaluating four GNN architectures (GCN, GAT, GIN, and AttentiveFP), selecting AttentiveFP as the primary model for its advanced attention mechanisms and edge feature integration. It employs a rigorous Bemis-Murcko scaffold split (80/10/10), utilises comprehensive 29-dimensional node and 12-dimensional edge features, and conducts Bayesian hyperparameter optimisation using Optuna, thereby addressing key methodological gaps in earlier EGFR GNN studies (Jiang et al. 2021).

2.7 Explainability in GNN-Based Drug Discovery

Despite their predictive power, deep learning models in drug discovery have historically been criticised as "black boxes," limiting their acceptance by medicinal chemists in practical settings (Luo, Y. Wang, et al. 2022). Post-hoc explainability techniques have been developed to bridge this gap. GNNExplainer (Ying et al. 2019) is a model-agnostic method that identifies the minimal subgraph (through node and edge masks) most responsible for a given prediction by maximising mutual information between the masked input and the model output.

Applied to EGFR inhibitors such as gefitinib, erlotinib, osimertinib, and lapatinib, GNNExplainer can highlight pharmacophoric features (e.g., hinge-binding nitrogens or hydrophobic halogens) in a chemically interpretable manner (Ying et al. 2019; Luo, Y. Wang, et al. 2022). While tools such as SHAP provide global feature importance, the current work focuses on GNNExplainer for instance-level insights that can be directly compared against known EGFR kinase pharmacophore models.

2.8 Virtual Screening with Machine Learning

Virtual screening (VS) is a cornerstone of computational drug discovery, enabling the prioritisation of large chemical libraries for experimental validation (M. Wang et al. 2023; Sun et al. 2021). In ligand-based machine learning VS, a model trained on known active and inactive compounds is deployed to score external libraries, ranking molecules by predicted probability of activity.

This thesis implements a focused VS pipeline by curating compounds active against related kinases (VEGFR2, BRAF, CDK2, ABL1, MET) from ChEMBL while strictly excluding the EGFR training set. After drug-likeness filtering (MW 250–600 Da, LogP -1 to 7), the optimised AttentiveFP model scores 1,872 unseen compounds. Top candidates are further prioritised using RDKit-computed ADMET properties, including Lipinski compliance and the Quantitative Estimate of Drug-likeness (QED) score (M. Wang et al. 2023; Tropsha, Muratov, et al. 2020).

Notable successes, such as the discovery of novel antibiotics through GNN-based screening of over 100 million compounds (Stokes et al. 2020), highlight the powerful potential of this approach. However, most published EGFR GNN models have stopped at classification without extending to prospective lead identification (Stokes et al. 2020; Malik, Sharma, and Kumar 2025).

2.9 Summary and Research Gap

The reviewed literature demonstrates substantial progress in EGFR-targeted therapy, bioactivity data curation, traditional cheminformatics, and GNN-based modelling. Nevertheless, critical gaps persist. Existing GNN studies for EGFR inhibitors typically lack one or more of the following: rigorous scaffold-based evaluation, advanced architectures with rich node/edge features, systematic interpretability using GNNExplainer on clinical compounds, and integration with targeted virtual screening pipelines for novel lead discovery (Malik, Sharma, and Kumar 2025; Jiang et al. 2021; Luo, Y. Wang, et al. 2022).

This thesis addresses these gaps through a comprehensive, end-to-end computational pipeline that combines high-quality ChEMBL curation, scaffold-split AttentiveFP modelling with Bayesian optimisation, GNNExplainer-based interpretability aligned with EGFR pharmacophores, and virtual screening of related-kinase inhibitors. By doing so, it contributes both methodological advancements in molecular machine learning and practical candidates for addressing drug-resistant NSCLC.

3. Materials and Methods

3.1 Overview of the Computational Pipeline

The overall methodology of this proposed study will follow a systematic, five-phase computational pipeline, as illustrated in Figure 1. This workflow is designed to progress from raw biochemical data curation to the identification of novel, drug-like lead compounds through advanced graph-based deep learning.

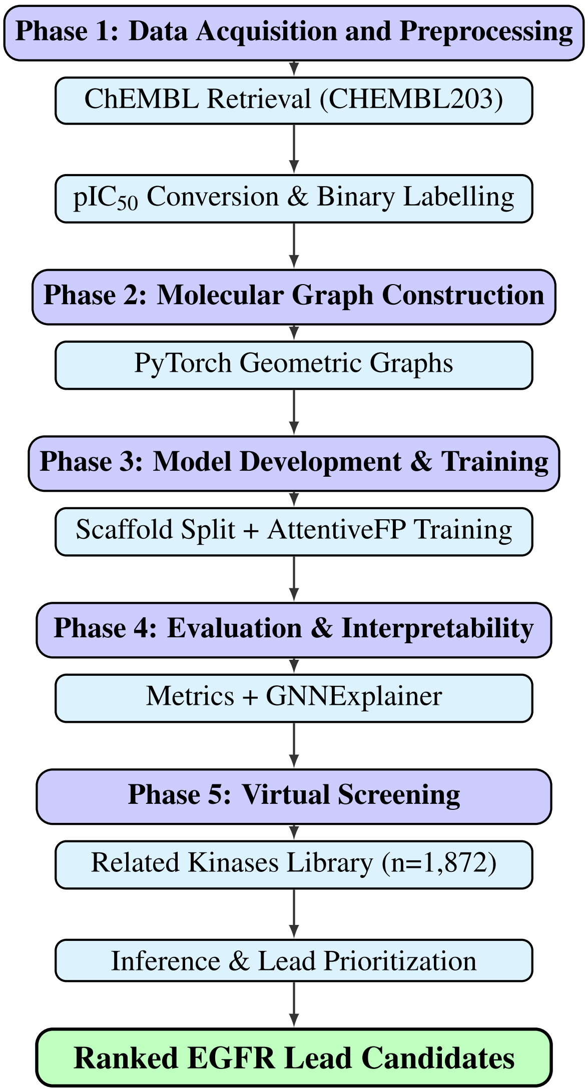

Figure 1: Proposed five-phase computational pipeline for GNN-based EGFR inhibitor classification and virtual screening.

3.2 Data Acquisition and Preprocessing

To build a robust predictive model, high-quality bioactivity data will be retrieved from the ChEMBL database (version 33 or later) via its RESTful API, specifically targeting the Epidermal Growth Factor Receptor (EGFR, ChEMBL ID: CHEMBL203). To ensure quantitative reliability, the dataset will be strictly filtered to include only records with exact Half-Maximal Inhibitory Concentration (IC_{50}) values, discarding any ambiguous measurements (e.g., '>', '<', or '$\sim$').

These values will be mathematically transformed into the negative logarithmic scale ($pIC_{50} = -\log_{10}(IC_{50} \times 10^{-9})$) to normalize the data distribution. A binary thresholding strategy will be implemented to frame the problem as a classification task: compounds with $pIC_{50} \geq 6.0$ (equivalent to $IC_{50} \leq 1\mu M$) will be classified as active inhibitors, while those below this threshold will be deemed inactive.

Rigorous structural standardization will be performed using RDKit's `MolStandardize` module. This pipeline will strip disconnected fragments (salts and solvents), neutralize charges where appropriate, and generate canonical SMILES to identify and remove duplicated entries. In cases where multiple bioactivity values exist for a single canonical SMILES, the median pIC_{50} will be retained. Finally, a physicochemical filter will isolate small molecules by restricting the molecular weight to a range of 100 to 1000 Daltons.

3.3 Molecular Representation and Graph Construction

Unlike traditional machine learning models that rely on fixed-length bit vectors (e.g., fingerprints), the proposed methodology will represent molecules natively as undirected graphs $\mathscr{G} = (\mathscr{V}, \mathscr{E})$, where $\mathscr{V}$ represents atoms (nodes) and $\mathscr{E}$ represents chemical bonds (edges). This transformation will be executed using PyTorch Geometric (PyG), enabling the preservation of complex topological and spatial relationships within the molecules.

3.3.1 Node (Atom) Features

Each node will be mathematically encoded as a comprehensive 29-dimensional continuous feature vector designed to capture fundamental biochemical properties essential for protein-ligand binding. The features will comprise:

- **Atom type** (one-hot encoded, 10 dimensions): C, N, O, S, F, Cl, Br, I, P, and Unknown.
- **Degree** (one-hot encoded, 6 dimensions): Capturing the number of covalently bonded heavy-atom neighbours (0 to 5).
- **Formal charge** (1 dimension): A critical parameter for modelling electrostatic interactions within the EGFR kinase pocket.

- **Attached hydrogens** (one-hot encoded, 5 dimensions): Ranging from 0 to 4, acting as a proxy for hydrogen bond donor capacity.
- **Hybridization state** (one-hot encoded, 5 dimensions): SP, SP2, SP3, SP3D, SP3D2, which dictates the local 3D geometry of the atom.
- **Aromaticity** (binary, 1 dimension): Crucial for identifying potential $\pi - \pi$ stacking interactions with aromatic residues in the binding site.
- **Ring membership** (binary, 1 dimension).

3.3.2 Edge (Bond) Features

To further enrich the message-passing framework, each edge will be encoded as a 12-dimensional feature vector:

- **Bond type** (one-hot encoded, 4 dimensions): Single, Double, Triple, Aromatic.
- **Conjugation** (binary, 1 dimension): Indicates extended electron delocalization.
- **Ring membership** (binary, 1 dimension).
- **Stereochemistry** (one-hot encoded, 6 dimensions): Captures E/Z and Cis/Trans configurations, which drastically affect ligand binding affinity.

3.4 Data Splitting Strategy

A major pitfall in cheminformatics machine learning is "scaffold memorization," where models perform exceptionally well on random test sets but fail to generalize to novel chemotypes. To rigorously evaluate the model's capacity for true virtual screening, a strict **Bemis-Murcko Scaffold Split** will be employed via RDKit.

The Bemis-Murcko algorithm systematically strips away terminal side chains, leaving only the core ring systems and linker atoms (the scaffold). The dataset will be partitioned based on these unique scaffolds into an 80% Training, 10% Validation, and 10% Test set. This guarantees that compounds sharing a core structural framework will be isolated into a single partition, forcing the model to evaluate entirely unseen molecular topologies during validation and testing.

3.5 Model Architectures

The core of this study will systematically evaluate four distinct Graph Neural Network (GNN) architectures functioning under the Message Passing Neural Network (MPNN) paradigm. The models to be developed include the Graph Convolutional Network

(GCN), Graph Attention Network (GAT), Graph Isomorphism Network (GIN), and AttentiveFP.

AttentiveFP is proposed as the primary, focal architecture for this research. Unlike standard GCNs that aggregate neighbor information isotropically, AttentiveFP utilizes a multi-step, graph-level attention mechanism. Crucially, it provides native support for integrating multidimensional edge features directly into the message-passing update equations. This capability is highly advantageous for molecular property prediction, as the specific nature of a chemical bond profoundly influences the electronic and spatial profile of a pharmacophore.

To establish a baseline and demonstrate the superiority of graph-based methods over established cheminformatics techniques, Random Forest (RF) and Logistic Regression models will also be trained. These baselines will utilize traditional 2048-bit ECFP4 (Morgan) fingerprints with a radius of 2 bonds, generated via RDKit.

3.6 Training and Hyperparameter Optimization

All GNN models will be trained using PyTorch. To address the anticipated class imbalance in the ChEMBL dataset (approximately 74% active compounds), a Weighted Binary Cross-Entropy (BCE) loss function with logits will be formulated. A positive class weight ($pos\_weight = 0.35$) will be applied to penalize misclassifications of the minority (inactive) class more heavily, ensuring the model does not bias its predictions toward the majority class.

The network weights will be optimized using the Adam optimizer. To ensure stable convergence, a `ReduceLROnPlateau` learning rate scheduler will be implemented (reduction factor = 0.5, patience = 10 epochs). An early stopping mechanism based on the Validation AUROC (patience = 20-30 epochs) will be utilized to prevent overfitting to the training scaffolds.

Hyperparameter optimization will be executed using the **Optuna** framework, leveraging the Tree-structured Parzen Estimator (TPE) algorithm for Bayesian optimization. The search space for the AttentiveFP model will be highly constrained, iterating over hidden dimensions, the number of GNN message-passing layers, the number of attention timesteps, dropout rates, batch sizes, learning rates, and L2 weight decay coefficients. The configuration yielding the highest AUROC on the validation set will be selected as the final model.

3.7 Evaluation Metrics

Due to the inherent class imbalance, relying solely on classification accuracy is insufficient and potentially misleading. Therefore, model performance will be assessed primarily using threshold-independent metrics:

- **Area Under the Receiver Operating Characteristic Curve (AUROC):** Measures the model's ability to distinguish between active and inactive classes across all possible classification thresholds.
- **Area Under the Precision-Recall Curve (AUPRC):** Heavily prioritizes the model's predictive precision specifically regarding the minority class, making it the most stringent metric for imbalanced data.

Secondary metrics, evaluated at a default probability threshold of 0.5, will include the F1-Score, the Matthews Correlation Coefficient (MCC)—which provides a balanced measure of true and false positives/negatives—and overall Accuracy. All final reported metrics will be calculated exclusively on the held-out Bemis-Murcko test set.

3.8 Model Interpretability

To overcome the "black-box" criticism frequently leveled at deep learning in drug discovery, **GNNExplainer** will be applied to the trained AttentiveFP model. This post-hoc explainability algorithm operates by maximizing the mutual information between the model's prediction and a compact subgraph structure.

GNNExplainer will generate node and edge importance masks, visually highlighting the specific atomic neighborhoods most influential in driving a positive prediction. Explanations will be deliberately generated for widely studied, clinically approved EGFR inhibitors (e.g., Gefitinib, Erlotinib, Osimertinib, and Lapatinib). The resulting computational attributions will be critically analyzed against established EGFR kinase pharmacophore models (verifying if the model successfully learned the importance of hinge-binding quinazoline nitrogens or hydrophobic halogen substituents). This step will validate that the model is learning genuine biochemical principles rather than spurious dataset correlations.

3.9 Virtual Screening Pipeline

The culmination of the study will involve deploying the optimized and validated AttentiveFP model in a prospective virtual screening pipeline. A specialized screening library will be curated by retrieving compounds reported as active against phylogenetically and structurally related kinases (e.g., VEGFR2, BRAF, CDK2, ABL1, and MET) from ChEMBL. Any compounds overlapping with the primary EGFR training set will be strictly excluded to guarantee external, zero-shot validation.

Prior to inference, the screening library will be filtered to ensure baseline drug-likeness. Only compounds with a molecular weight between 250 and 600 Da, and a calculated LogP between -1 and 7, will be retained. The AttentiveFP model will then perform forward-pass inference on this unseen library (estimated $n = 1,872$), generating a predicted probability of EGFR inhibition for each compound.

To triage the highest-scoring candidates for potential future experimental validation, a suite of ADMET-proxy properties will be calculated using RDKit. These will include Molecular Weight (MW), calculated partition coefficient (LogP), the number of hydrogen bond donors and acceptors, Topological Polar Surface Area (TPSA), strict compliance with Lipinski's Rule of Five, and the Quantitative Estimate of Drug-likeness (QED) score. Candidates exhibiting both a high predictive probability (> 0.85) and an optimal QED score will be proposed as high-value, novel lead compounds.

3.10 Software and Implementation Details

All computational modeling and data processing will be performed using open-source Python (version 3.10+) libraries to ensure complete reproducibility. Core packages will include RDKit (2023.09+) for molecular standardization and descriptor generation, PyTorch and PyTorch Geometric for GNN construction and training, Optuna for Bayesian hyperparameter tuning, and scikit-learn for baseline modeling and metric computation. Visualizations, including GNNExplainer outputs, will be generated using Matplotlib and Seaborn. All deep learning experiments will be executed on GPU-accelerated hardware (NVIDIA CUDA architecture) to facilitate the computationally intensive training of graph neural networks.

Expected Outcomes

The expected outcomes of this study are directly aligned with the specific objectives and research questions outlined in Chapter 1. Upon successful completion of the proposed research, the following deliverables will be generated:

- Fully trained and hyperparameter-optimised Graph Neural Network models (specifically AttentiveFP, GCN, GAT, and GIN) for the binary classification of EGFR inhibitors, with predictive performance rigorously reported using AUROC, AUPRC, accuracy, F1-score, and MCC under a strict Bemis-Murcko scaffold split.

- A comprehensive benchmarking analysis comparing the performance of the advanced GNN architectures against traditional cheminformatics baselines (Random Forest and Logistic Regression using ECFP4 fingerprints), alongside an architectural and hyperparameter sensitivity analysis for the primary AttentiveFP model.

- GNNExplainer atom-level and bond-level importance visualisations for clinically relevant EGFR inhibitors (e.g., gefitinib, erlotinib, osimertinib, and lapatinib). This will include a written chemical interpretation comparing the model's learned structural attributions to established EGFR kinase pharmacophores from the medicinal chemistry literature.

- A ranked virtual screening hit list of potential novel EGFR inhibitor candidates identified from the unseen library of related-kinase inhibitors (n=1,872). This list will include predicted activity probability scores, computationally derived ADMET-proxy properties (e.g., Molecular Weight, LogP, Lipinski compliance, and QED score) to assess drug-likeness, and GNNExplainer visualisations for the top-priority hits.

- A fully reproducible, open-source Python computational pipeline covering all phases of the study (data curation $\rightarrow$ PyG graph construction $\rightarrow$ model training with Optuna optimisation $\rightarrow$ explainability and virtual screening). This pipeline will be suitable for adaptation to other kinase drug discovery targets.

Work Plan of the Study

The proposed time frame and activities of the study are presented below.

Table 1: Time Schedule

Tasks	**Weeks**					
	1	**2**	**3**	**4**	**5**	**6**
Literature Review						
Data Collection and Preprocessing						
Model Development and Training						
Model Evaluation and Analysis						
Report Writing						
Final Defense						

References and Citations

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